

UNIVERSITY OF BUEA

**FACULTY OF ENGINEERING
AND TECHNOLOGY**

**DEPARTMENT OF
COMPUTER ENGINEERING**

**A DISTRIBUTED BLOCKCHAIN-BASED APPROACH TO
PROTECT IDENTITY, INTEGRITY AND PRIVACY IN
AGRICULTURAL DATA PROCESSING**

By

FORCHA GLEN BELOA
(FE22P049)

M.Eng. in Telecommunications and Networks

A Thesis submitted to the Faculty of Engineering and Technology of the
University of Buea in partial fulfilment of the requirements
for the degree of Doctor of Philosophy (PhD)
in Computer Engineering.

Supervisors

Michael Ekonde SONE, PhD
Associate Professor

Elie FUTE TAGNE, PhD
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August 2026

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DEDICATION

To ...

CERTIFICATION

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The Thesis of **FORCHA GLEN BELOA** (FE22P049) entitled “**A Distributed Blockchain-Based Approach to Protect Identity, Integrity and Privacy in Agricultural Data Processing**” submitted to the Department of Computer Engineering, Faculty of Engineering and Technology of the University of Buea in Partial Fulfilment of the Requirements for the Award of the Doctor of Philosophy (PhD) Degree in Computer Engineering under the Supervision of

Sign:_____

Date:_____

Michael Ekonde SONE, PhD
Associate Professor
(Supervisor)

Sign:_____

Date:_____

Elie FUTE TAGNE, PhD
Associate Professor
(Supervisor)

Sign:_____

FORCHA GLEN BELOA

ACKNOWLEDGEMENTS

ABSTRACT

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LIST OF ABBREVIATIONS

ABAC	Attribute-Based Access Control
ACK	Acknowledgement
ACID	Atomicity, Consistency, Isolation, Durability
ADC	Analogue-to-Digital Converter
API	Application Programming Interface
ARM64	64-bit Advanced RISC Machine architecture
BATMAN-adv	Better Approach To Mobile Ad-hoc Networking, advanced
BLE	Bluetooth Low Energy
BFT	Byzantine Fault Tolerance
CA	Certificate Authority
CAP	Consistency, Availability, Partition tolerance
CFT	Crash Fault Tolerance
CPU	Central Processing Unit
CRT	Chinese Remainder Theorem
DID	Decentralised Identifier
DLT	Distributed Ledger Technology
DP	Differential Privacy
ECDSA	Elliptic Curve Digital Signature Algorithm
EDP	Energy-Delay Product
ESP32	Espressif 32-bit microcontroller with integrated Wi-Fi and Bluetooth
ETSI	European Telecommunications Standards Institute
GDPR	General Data Protection Regulation
GW	Gateway
HLF	Hyperledger Fabric
HRBAC	Hierarchical Role-Based Access Control
IoT	Internet of Things
IPFS	InterPlanetary File System
JSON	JavaScript Object Notation

LoRa	Long Range (sub-GHz spread-spectrum radio modulation)
LoRaWAN	Long Range Wide Area Network
LPWAN	Low-Power Wide-Area Network
LZW	Lempel-Ziv-Welch compression
MAC	Medium Access Control
MCU	Microcontroller Unit
MSP	Membership Service Provider
PBFT	Practical Byzantine Fault Tolerance
PKI	Public Key Infrastructure
RAM	Random Access Memory
RBAC	Role-Based Access Control
RF	Radio Frequency
RNS	Residue Number System
ROI	Return on Investment
RPi	Raspberry Pi
RTC	Real-Time Clock
SF	Spreading Factor
SHA	Secure Hash Algorithm
SLO	Service Level Objective
SSS	Shamir Secret Sharing
TEE	Trusted Execution Environment
TPS	Transactions Per Second
TPM	Trusted Platform Module
UTXO	Unspent Transaction Output
WSN	Wireless Sensor Network
X.509	ITU-T public-key certificate standard
ZKP	Zero-Knowledge Proof

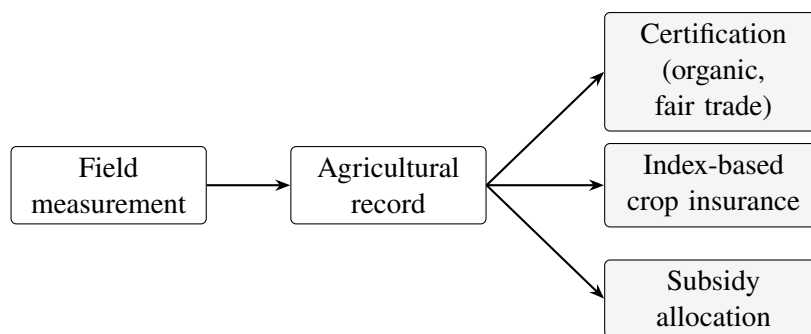
CHAPTER ONE

INTRODUCTION

1.1. Background to the Study

Agriculture is undergoing a transition from experience-driven practice to data-driven decision making. Networks of low-cost sensors now measure soil moisture, canopy temperature, humidity and nutrient status continuously, and the resulting streams feed irrigation controllers, yield models and certification schemes [1, 2]. In water-stressed regions this shift is not a convenience but a necessity. Irrigation accounts for the dominant share of freshwater withdrawal, and climate variability has compressed the margin for error in scheduling decisions [3, 4].

A measurement taken in a field does not stop at the field. It becomes a record, and that record is consumed by parties with interests of their own, as Figure 1 sets out. The value of the record to those parties is exactly what creates the incentive to falsify it, and the further a record travels from the instrument that produced it, the weaker the evidence that it is genuine.



*Each downstream use attaches value to the record,
and therefore creates an incentive to falsify it.*

Figure 1. Lifecycle of an agricultural measurement: a reading becomes a record whose downstream value creates the incentive to falsify it.

Reviews of the sensing layer converge on the same reading. Smart sensing has matured to the point where measurement is no longer the constraint, while the integration of

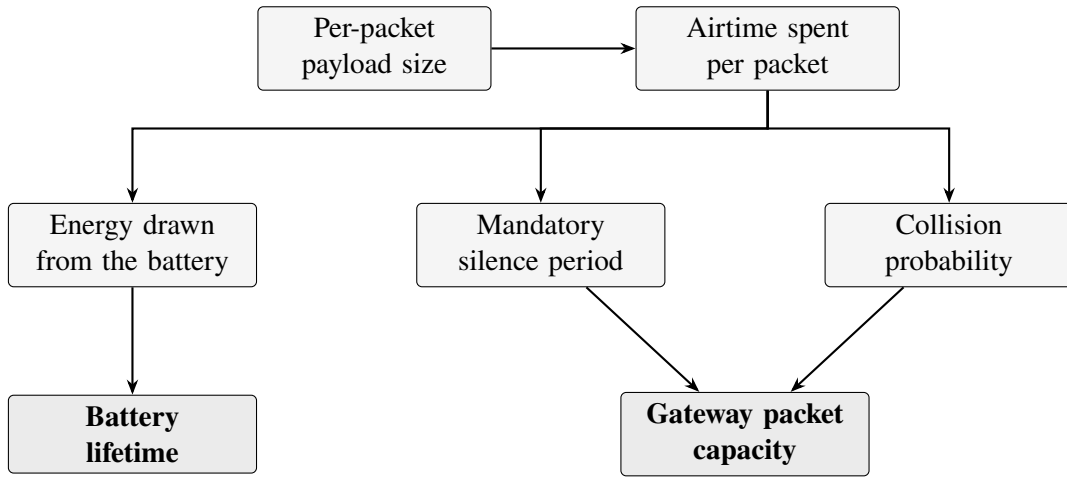
that measurement into auditable decision-making remains [5, 6]. Surveys of edge-enabled agriculture name communication reliability and energy constraints as the two bottlenecks that most often defeat field deployments [6], and surveys of the wider IoT threat surface agree that the constrained endpoint is the weakest link in the provenance chain [7, 8].

The economics of the sensing layer explain why the problem is hard rather than merely tedious. A deployment that is worth installing on a smallholding must cost a small fraction of the crop value it protects, which forces the node bill of materials down to a level at which neither a hardware security module nor a generous energy budget is affordable. The same economics rule out mains power and wired backhaul across a field, so the design space collapses onto battery-powered microcontrollers speaking a long-range, low-rate radio. Every protection mechanism proposed in this thesis has to survive inside that envelope, and a mechanism that works only on a mains-powered gateway has solved the easier half of the problem.

The sensing layer of such systems is built from severely constrained devices. An Espressif 32-bit microcontroller with integrated Wi-Fi and Bluetooth (ESP32) node operating on battery and harvested solar power must transmit over kilometre-scale distances using Long Range (sub-GHz spread-spectrum radio modulation) (LoRa), a modulation scheme that trades data rate for range and therefore charges a high airtime cost per packet [9, 10]. Airtime is the binding constraint. It determines energy consumption at the node, and, through duty-cycle regulation and collision probability, it determines how many nodes a single gateway can serve [11]. Conventional responses compress the payload, but general-purpose compressors such as Huffman and Lempel-Ziv-Welch compression (LZW) carry dictionary and buffer overheads that a microcontroller with a few kilobytes of free Random Access Memory (RAM) can ill afford [12, 13].

Two further constraints sharpen this. Sub-GHz IoT bands are governed by a

duty-cycle limit, so a node that has just transmitted must stay silent for a period proportional to the airtime it consumed. Airtime is therefore not only an energy cost but a rate limit: it caps how often a node may report, and no increase in battery capacity relaxes it. Second, because LoRa uses unslotted access, packets collide whenever two nodes transmit on the same channel at the same time, and the collision probability grows with the square of offered traffic in the regime that matters. Halving per-packet occupancy therefore buys more than a proportional reduction in collisions, which is why the mechanism studied here is worth the arithmetic it costs. Figure 2 sets out how one quantity, airtime per packet, propagates into three consequences that a design cannot address independently.



A shorter payload lowers airtime once, and that single reduction pays three times: less energy drawn per reading, a shorter mandatory silence period, and a lower collision probability at the gateway.

Figure 2. Airtime is a rate limit, not only an energy cost: reducing it relaxes the duty-cycle and collision ceilings on gateway capacity.

The Chinese Remainder Theorem (CRT) offers a different route. Decomposing a reading into residues modulo pairwise coprime moduli yields several short packets in place of one long one [14, 15]. Because a multi-channel concentrator such as the SX1302 demodulates channels concurrently, the short residue packets can be received simultaneously, and the original value is recovered in $O(k)$ operations for k moduli. The practical effect is a reduction in per-packet occupancy rather than in total transmitted

volume, which is precisely the quantity that governs gateway capacity. The detailed encoding and reconstruction path is presented in Chapter Three.

Residue arithmetic is not new to distributed systems. The same decomposition underlies the throughput work in permissionless ledgers, where partitioning a workload across independent classes lets operations commit concurrently that would otherwise serialise [16–18], and the algebraic machinery itself has been carried well beyond integer congruences [19]. That line of work establishes the arithmetic and the parallelisation argument. What it does not address is the constrained radio link, where the payoff is airtime rather than processor cycles, nor the identity and confidentiality questions that arise when the residues cross a shared medium. A single result hints that the second is reachable: residue decomposition combined with chaotic maps has been used to conceal image content reversibly [20], which suggests residues can carry confidentiality as well as efficiency, albeit under a different medium and threat model. This thesis takes the arithmetic as established and asks what it is worth on the edge of the network rather than at its centre [21].

Data that has been collected cheaply must then be trusted. Agricultural records increasingly carry consequences beyond the farm gate. They substantiate organic and fair-trade certification, they underwrite index-based crop insurance, and they inform subsidy allocation. Each of these uses creates an incentive to falsify. A centralised database concentrates that risk in a single administrator, and an unprotected sensor network invites the injection of fabricated readings by any party with a compatible radio. Distributed Ledger Technology (DLT) addresses the tamper-evidence requirement by binding records into a hash-linked chain replicated across mutually distrusting parties, and permissioned platforms such as Hyperledger Fabric (HLF) add an identity layer through a Membership Service Provider (MSP) and Public Key Infrastructure (PKI), making it feasible to say not

merely that a record is unaltered but who produced it [22–24].

Running that machinery at the edge is the difficulty. Consensus, endorsement and world-state maintenance were designed for datacentre hardware, whereas an agricultural deployment must place gateways in the field, on 64-bit Advanced RISC Machine architecture (ARM64) single-board computers, behind intermittent connectivity. Whether a permissioned ledger can meet irrigation-control timing requirements on such hardware has until recently been asserted rather than measured.

Finally, protection is not exhausted by integrity. A soil-moisture trace is also a disclosure. It reveals plot productivity, irrigation strategy and, in aggregate, the commercial position of the farmer. Where records are replicated across a consortium that includes buyers, insurers and cooperatives, the same replication that secures integrity enlarges the audience for private information. Identity, integrity and privacy therefore have to be treated as one design problem rather than three independent ones, which is the position this thesis takes.

1.2. Statement of the Problem

Three properties that ought to hold simultaneously are, in current systems, traded against one another.

Identity is weak at the sensing layer. Constrained nodes seldom carry credentials strong enough to authenticate individual readings. Where authentication exists it typically terminates at the gateway, so the ledger attests to what the gateway submitted rather than to what the sensor measured. The provenance chain has a gap exactly where falsification is cheapest, at the final hop between an individual node and the record that claims to represent it.

Farm-specific work confirms that the exposure is real rather than theoretical. Tampering with IoT farm data has been treated as a distinct security problem requiring its

own countermeasures [25], and production deployments in food chains report provenance as the primary motivation for adopting a ledger at all [26, 27].

Integrity guarantees do not survive the move to the edge. Permissioned ledgers provide tamper evidence, but their published performance characterisations assume server-class hosts. On ARM64 gateways with constrained memory and intermittent links, it is not established that endorsement and ordering can complete inside the control loop of an irrigation decision, nor how the ledger behaves during network partition and recovery.

Privacy is sacrificed to obtain integrity. Replicating raw agronomic readings to every consortium member is the simplest way to make them verifiable and the worst way to keep them confidential. Existing deployments largely accept this trade, or push confidential data off chain in a way that reintroduces the trusted intermediary the ledger was meant to remove.

The three shortfalls, shown together in Figure 3, are not independent faults to be repaired one at a time. Each follows from the same scarcity, and a remedy applied to one is paid for out of the budget available to the others.

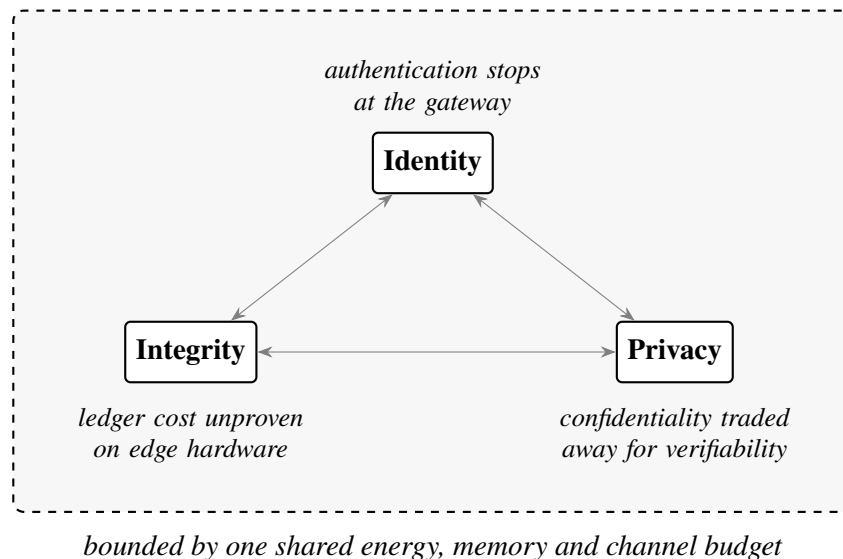


Figure 3. The three protection properties, sharing one constrained budget and therefore one design problem.

Underlying all three is a resource problem. The transmission budget that would

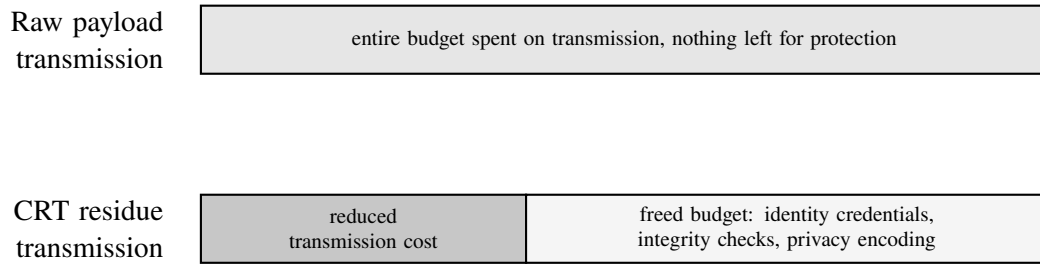
pay for per-reading signatures, encryption and privacy-preserving encodings is already consumed by the raw payload itself. Any scheme that adds cryptographic protection without first reducing transmission cost will exhaust the node's energy budget or the gateway's channel capacity.

How, then, can identity, integrity and privacy be protected simultaneously in an agricultural data processing pipeline, end to end from constrained sensing node to distributed ledger, within the energy, memory and channel-capacity budgets that field deployment actually permits?

1.3. Rationale of the Study

The budget for protection has to come from somewhere. If CRT residue decomposition reduces per-packet airtime substantially, it frees channel capacity and node energy that can be reinvested in protection rather than merely banked as efficiency. The same decomposition has a second, less obvious property. An individual residue discloses little about the original value, so residues distributed across channels or custodians provide a natural substrate for confidentiality and for threshold reconstruction. A mechanism introduced for efficiency can thus be made to serve privacy, which is what makes the combination worth studying rather than treating transmission and security as separate layers. Figure 4 makes the reallocation concrete: the same fixed budget that raw transmission spends in full is only partly spent once residues replace it, and the remainder is what pays for protection.

The two mechanisms also compose cleanly. Residue decomposition and ledger verification operate on the same object at different points in its life, the first on the reading in flight and the second on the record at rest, and the quantity each is sensitive to is different: the radio cares about occupancy, the ledger about the size and count of committed transactions. A scheme that reduces occupancy without increasing transaction



The budget that raw transmission consumes in full is only partly spent once CRT decomposition lowers per-packet airtime; the freed remainder is what pays for the identity, integrity and privacy mechanisms Chapter Three adds.

Figure 4. Reallocating the transmission budget: airtime saved by residue decomposition funds identity, integrity and privacy protection.

count therefore improves one without taxing the other, which is not generally true of protection mechanisms and is worth demonstrating rather than assuming.

There is also an evidential gap. Claims about edge blockchain feasibility are typically supported by simulation or short laboratory runs. A sustained field deployment on real hardware, through real weather and real connectivity failures, produces a different and more useful class of evidence, including the failure modes that simulation does not generate.

The applied case is direct. Smallholder and arid-region agriculture is where both the water-efficiency gains and the certification-fraud exposure are largest, and where infrastructure assumptions of existing systems are least likely to hold.

1.4. Objectives of the Study

1.4.1. Main Objective

To design, implement and evaluate a distributed blockchain-based framework that protects identity, integrity and privacy in agricultural data processing, and to validate it in a sustained field deployment on constrained edge hardware.

1.4.2. Specific Objectives

1. **Analyse** the identity, integrity and privacy requirements of agricultural data pipelines, decomposing the threat surface from sensing node through gateway to ledger into an attacker model with stated capabilities and assumptions.
2. **Design** an integrated protection scheme combining CRT multi-channel residue transmission, which lowers per-packet airtime while preserving exact reconstruction, with an identity layer that binds each reading to an authenticated originating node through HLF MSP credentials and HRBAC policy enforced in chaincode.
3. **Implement** the framework on a heterogeneous edge cluster of ESP32 sensing nodes and ARM64 Raspberry Pi (RPi) gateways running HLF under Raft ordering, deployed under real agricultural operating conditions.
4. **Evaluate** the transmission, ledger and access-control layers against compression and centralised baselines on airtime, energy, throughput, latency and memory footprint, and assess the scalability and fault-tolerance envelope through queuing-theoretic analysis validated against measured load.

1.5. Research Questions

- RQ1.** To what extent can CRT residue decomposition reduce per-packet airtime and increase gateway packet capacity without loss of reconstruction fidelity?
- RQ2.** Can a permissioned distributed ledger sustain the throughput and latency required by real-time irrigation control when hosted entirely on ARM64 edge gateways?
- RQ3.** What identity binding is required for a ledger record to attest to the originating sensing node rather than merely to the submitting gateway, and what does that

binding cost?

RQ4. What confidentiality is obtained by distributing residues across channels and custodians, and under what adversary model does it hold?

RQ5. How does the integrated framework behave as offered load approaches gateway and ledger saturation, and where are its practical limits?

1.6. Research Hypotheses

H1 (Transmission efficiency). CRT residue transmission reduces per-packet airtime by at least 40% relative to raw transmission while maintaining at least 99% reconstruction accuracy.

H2 (Edge ledger performance). HLF hosted on four RPi 4B gateways sustains a mean transaction latency below 2 s at an average throughput of at least 50 TPS.

H3 (Access-control overhead). Enforcing Role-Based Access Control (RBAC) in chaincode preserves at least 50 TPS and keeps mean smart-contract execution below 500 ms.

H4 (Field robustness). Sustained field operation maintains at least 99% end-to-end reliability, with measured airtime within 5 ms of the analytical model.

H5 (Privacy preservation). An adversary observing fewer than k of the k residue channels cannot reconstruct the original reading, and the residual information leakage remains below a stated bound.

1.7. Scope and Delimitations

In scope. Sub-GHz LoRa sensing networks of the order of fifty nodes; permissioned ledgers of the HLF family under Raft ordering; gateway hardware in the RPi 4B class;

agronomic telemetry, specifically soil moisture, temperature, humidity and irrigation actuation; and the identity, integrity and privacy properties of that pipeline.

Out of scope. Public permissionless blockchains and their consensus economics; cellular and satellite backhaul; image-based and remote-sensing agronomy; agronomic model accuracy as such, which is treated as given rather than as an object of study; and formal cryptographic proofs of the underlying primitives, which are cited rather than re-derived.

Delimitations. Field validation is drawn from a single deployment site and a single cropping season, which bounds the claims that can be made about generalisation across soil types and climates. Node population is bounded by what could be instrumented. Behaviour beyond that population is established analytically and by synthetic load rather than by direct measurement.

1.8. Significance of the Study

For **engineering practice**, the work supplies measured rather than simulated evidence about what a permissioned ledger costs on edge hardware, and a transmission scheme that reduces that cost enough to make protection affordable.

For **the research literature**, it connects two bodies of work that have developed separately, namely residue arithmetic for communication efficiency and distributed ledgers for data provenance, and shows that the first can subsidise the second.

Methodologically, the work contributes a characterisation obtained under field conditions rather than in a laboratory. Failures that a testbed does not generate, mesh partitions, thermal derating, intermittent backhaul and the recovery behaviour that follows each, are exactly the ones that determine whether such a system is deployable, and they only appear when the system is left running.

For **agricultural stakeholders**, it offers a path to verifiable certification and

insurance records that does not require farmers to surrender commercially sensitive detail to every consortium member.

1.9. Definition of Key Terms

Airtime The time a radio occupies a channel to transmit one packet. It governs both node energy consumption and gateway capacity.

Residue The remainder of a reading modulo one of a set of pairwise coprime moduli. A complete residue set uniquely determines the original value below the product of the moduli.

Integrity The property that a record has not been altered since it was committed, and that alteration would be detectable.

Identity The property that a record can be attributed to a specific authenticated principal, here a specific sensing node rather than merely a gateway.

Privacy The property that a record's content is disclosed only to principals authorised to see it, notwithstanding replication of the ledger.

Endorsement In HLF, the process by which designated peers simulate a transaction and sign the result before ordering.

1.10. Organisation of the Thesis

Chapter One states the background, problem, rationale, objectives, questions and hypotheses. **Chapter Two** reviews the literature on precision agriculture and IoT sensing, transmission optimisation, distributed ledgers in agriculture, edge-resident consensus and privacy-preserving data processing, and identifies the gap this work addresses. **Chapter Three** presents the materials and methods: system architecture, CRT formulation, identity

and access-control design, hardware implementation, deployment protocol and evaluation methodology. **Chapter Four** reports results and applications across the transmission, ledger, access-control and agricultural-outcome dimensions. **Chapter Five** discusses the findings against the hypotheses, states limitations and threats to validity, and sets out conclusions, recommendations and perspectives for further research.

CHAPTER TWO

REVIEW OF THE LITERATURE

2.1. Introduction

Three lines of research, developed largely apart from one another, converge on the problem we address.

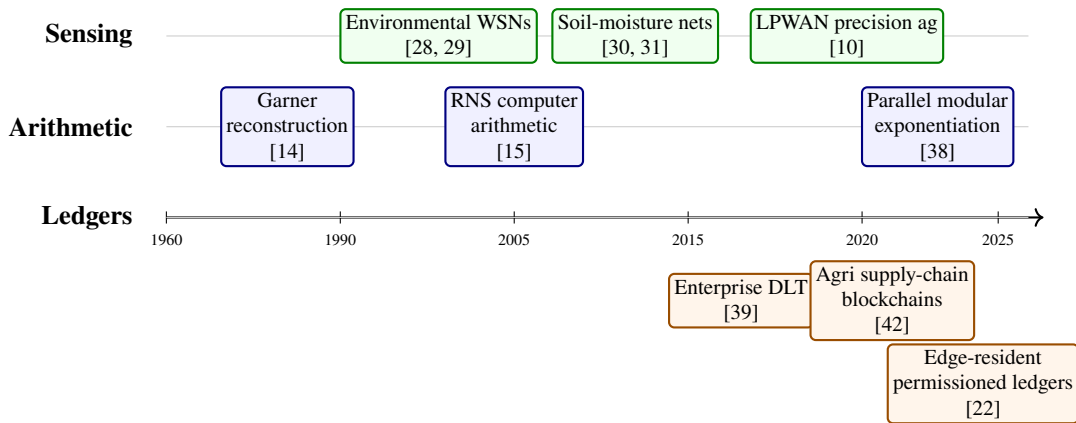
The first is the instrumentation of agriculture itself. Wireless sensor networks for environmental monitoring emerged in the early 2000s as a low-cost alternative to wired instrumentation, with early deployments demonstrating that battery-powered nodes could report soil and microclimate variables over months without maintenance [28, 29]. Soil-moisture networks matured through the 2000s and 2010s into standard tools for irrigation scheduling [30, 31], and the arrival of Low-Power Wide-Area Network (LPWAN) radios after 2015 extended the useful range of such networks from tens of metres to several kilometres, at the cost of a severely constrained airtime budget [10, 32]. Precision agriculture as a discipline absorbed this instrumentation gradually, and by the early 2020s the technical barrier to dense sensing had largely fallen, while the barrier to trustworthy, verifiable use of the resulting data had not [33, 34].

The second lineage is arithmetic rather than agricultural. The Chinese Remainder Theorem states a result on simultaneous congruences known in essentially its modern form since antiquity, formalised for modern computation by Garner’s constructive reconstruction procedure [14] and developed since as the mathematical basis of residue number systems for fast, parallel, fault-tolerant arithmetic [15, 35]. That literature has run for decades inside computer arithmetic and cryptography, largely untouched by networking or agriculture, its natural home being hardware multipliers, error-correcting codes and, more recently, the parallel verification of cryptographic signatures [36–38].

The third lineage is distributed ledgers. Blockchain moved from a cryptocurrency

mechanism to a general tamper-evident data structure once enterprises recognised that a hash-linked, replicated log could substitute for a trusted central database among mutually distrusting parties [39, 40]. Permissioned platforms such as Hyperledger Fabric formalised that idea for consortium settings, adding an identity layer through a MSP [23, 41], and agricultural supply chains were an early, natural application because certification and provenance are exactly the tamper-evidence problem the ledger solves [42–44].

Figure 5 places these three lineages on one timeline. They do not meet until the last five years, and even where they meet, as the sections that follow show, they meet pairwise rather than as a single design. The rest of this chapter lays the foundations in the order a reader needs them: what a blockchain actually is and why a permissioned one suits a consortium of mutually distrusting parties (Section 2.2(a)), the agricultural sensing and transmission layer that produces the data (Sections 2.2(b)-2.2(c)), the residue arithmetic we use to decompose readings with (Section 2.2(d)), how blockchain has actually been used in agriculture (Section 2.3(a)), and the consensus, identity and scalability literature that determines what a ledger costs to run at the edge (the sections after that).



*The three lineages converge only in the present decade,
and, as the sections below show, pairwise rather than as one design.*

Figure 5. Three lineages underlying our work, developed independently until converging in the 2020s.

2.2. Foundations and Enabling Technologies

The four strands below supply the technical vocabulary the rest of the chapter uses: what a distributed ledger guarantees and at what cost, how agricultural sensing is instrumented, what the radio charges for a packet, and what residue arithmetic offers.

a) Fundamentals of Blockchain Technology

Before turning to agriculture specifically, we set out what a blockchain is, because the word covers a wide family of designs and the choice among them is not cosmetic. A blockchain is, at its simplest, an append-only ledger replicated across a set of nodes that do not fully trust one another, ordered into blocks that are cryptographically chained together [39, 45]. Each block B_i commits to its predecessor and its own transaction set through

$$h_i = H(h_{i-1} \parallel \text{tx}_i), \quad (2.1)$$

where H is a cryptographic hash function, so that altering any committed tx_j for $j \leq i$ requires recomputing every $h_j, \dots, h_i$ across every replica that holds the chain, a cost that grows with the number of mutually distrusting replicas rather than shrinking [46]. Merkle's original construction generalises Equation 2.1 to a binary tree of hashes so that membership of a single transaction can be proven in $O(\log n)$ rather than by revealing the whole block, a property later systems use for compact audit proofs [47]. This hash-chaining property, not any particular cryptocurrency, is what the literature means by tamper evidence, and it is what agricultural applications actually draw on: a record, once committed and replicated, cannot be silently rewritten by any single party.

Blockchains split first on who may participate. Permissionless (public) chains such as Bitcoin and Ethereum admit any node and settle disagreement through open,

resource-costly consensus, typically proof of work, so that attacking the chain costs more than any single participant can afford [48, 49]. Permissioned (private or consortium) chains restrict participation to identified, vetted members, replacing open competition with a membership service that binds every submitted transaction to an accountable identity [23, 50]. Hyperledger Fabric, the platform we deploy, is representative of the permissioned family: a MSP issues ITU-T public-key certificate standard (X.509) certificates to every organisation and peer, an execute-order-validate pipeline separates transaction endorsement from ordering, and chaincode, Fabric’s term for a smart contract, encodes the business logic that peers execute and validate before a transaction is committed [23, 51]. A permissionless design buys resistance to an unknown, potentially adversarial population of participants at the cost of energy and finality latency; a permissioned design accepts a smaller, vetted membership in exchange for finality in seconds rather than minutes and no mining cost, which is why every agricultural deployment reviewed in this chapter, without exception, chooses the permissioned family. A consortium of farms, certifiers and buyers is exactly the setting permissioned platforms are built for: participants who must cooperate on shared records without extending one another unconditional trust, and who are identifiable and accountable rather than anonymous [45, 52].

Smart contracts are the mechanism that turns a passive ledger into an active one. Once deployed, chaincode executes deterministically on every endorsing peer, so that business rules, who may submit a reading, what counts as a valid range, how a certification decision is recorded, are enforced by consensus over code rather than by a single administrator’s discretion [27, 50]. Alternative ledger topologies exist outside the block-and-chain model proper, directed-acyclic-graph ledgers such as IOTA dispense with blocks altogether and let each new transaction validate two prior ones [53], trading the strict global ordering a blockchain provides for higher throughput under light load; we

return to this trade-off, and to consensus proper, in Section 2.3(b). What every variant shares, and what the rest of this chapter takes as given, is the replicated, hash-chained record that makes a committed reading expensive to falsify after the fact. Underlying every transaction's authenticity, in both permissioned and permissionless designs, is public-key cryptography: a node signs what it submits with a private key, and peers verify that signature against a public one before accepting it, typically an elliptic-curve scheme such as Elliptic Curve Digital Signature Algorithm (ECDSA) over the secp256k1 curve [49]. Signature verification is itself a parallelisable modular-arithmetic computation, which is the property Section 2.2(d) returns to when it discusses residue number systems.

b) Precision Agriculture and Internet of Things Sensing

Precision agriculture research has consistently found that sensing density and irrigation efficiency move together. IoT soil and canopy sensing networks demonstrated early that automated, sensor-driven scheduling outperforms calendar-based irrigation on both water use and yield [1, 2, 54, 55], a finding replicated across arid and semi-arid contexts where the water-saving incentive is largest [3, 4, 56]. Reviews of the field converge on the same three limitations. First, decision-support integration remains fragmented: sensor data reaches a dashboard but rarely a closed-loop controller with an auditable rationale [33, 34, 57, 58]. Second, most systems evaluate the analytics rather than the network that feeds them, so machine-learning yield and stress models are reported against clean, simulated input streams rather than the loss and jitter that a field deployment actually produces [59–62]. Third, sensing accuracy in the field degrades in ways a laboratory calibration does not capture, drift, fouling and intermittent failure among them, and few studies report the resulting error budget [63–66]. Later generations of this literature push toward higher automation, describing four- and five-generation smart-farming stacks that

assume the identity and integrity of every reading rather than establishing it [67–71]. That assumption is exactly the gap Section 2.3(a) returns to.

c) Low-Power Wide-Area Network Transmission Optimisation and Energy Modelling

LoRa is the dominant physical layer in this literature because it reaches farm-scale distances, of the order of kilometres in rural terrain, from a battery node at sub-milliwatt receive power [9, 32, 72]. The cost is airtime. The Semtech chirp spread-spectrum design fixes the time to transmit a payload of PL bytes as a function of spreading factor SF, bandwidth BW, coding rate CR and the low-data-rate optimisation flag DE [73, 74]. The symbol period is

$$T_{\text{sym}} = \frac{2^{\text{SF}}}{\text{BW}}, \quad (2.2)$$

the payload requires

$$n_{\text{payload}} = 8 + \max\left(\left\lceil \frac{8 PL - 4 \text{SF} + 28 + 16 \text{CRC} - 20 \text{IH}}{4(\text{SF} - 2 \text{DE})} \right\rceil (\text{CR} + 4), 0\right) \quad (2.3)$$

symbols, and total airtime is

$$T_{\text{air}} = \underbrace{(n_{\text{preamble}} + 4.25)}_{\text{preamble}} T_{\text{sym}} + n_{\text{payload}} T_{\text{sym}}. \quad (2.4)$$

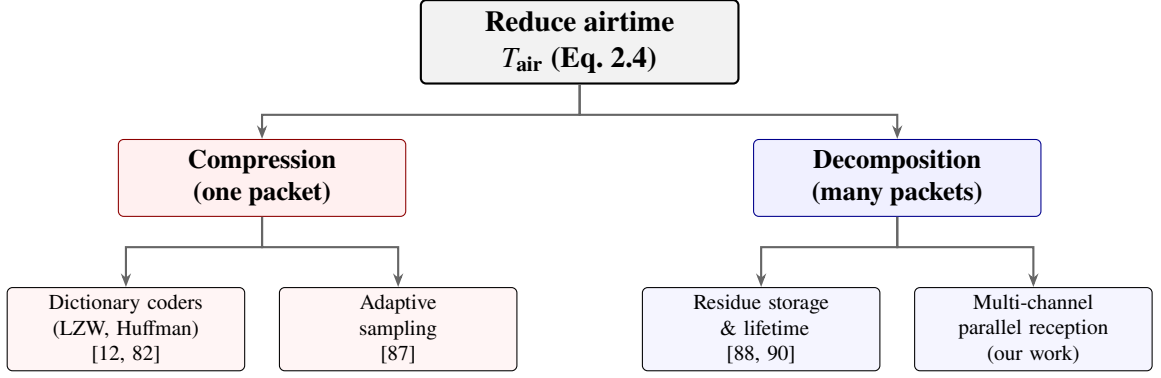
Equation 2.4 is the quantity every optimisation in this literature ultimately reduces: for a fixed SF and BW, airtime is affine in payload length, so a shorter payload is the only lever available once the modulation is chosen. Regulatory duty-cycle limits then convert that airtime into a rate ceiling rather than merely an energy cost. Under European Telecommunications Standards Institute (ETSI) EN 300 220, a sub-GHz node may occupy

its channel for at most a fixed fraction $DC_{\max}$ of any observation window [75, 76], so a single reading consumes

$$DC = \frac{T_{\text{air}}}{T_{\text{window}}} \leq DC_{\max}, \quad (2.5)$$

and offered channel load across N nodes reporting once per window is $G = N T_{\text{air}}/T_{\text{window}}$ Erlang, which under the pure-ALOHA assumption common to this literature bounds per-packet collision probability at $P_{\text{coll}} = 1 - e^{-2G} \approx 2G$ for small G [10, 74]. Node energy per transmission follows directly, $E_{\text{tx}} = P_{\text{tx}} T_{\text{air}}$, so any reduction in T_{air} is simultaneously an energy saving and a capacity gain, which is why the transmission-optimisation literature and the energy-modelling literature are, in practice, one body of work [11, 77–81].

Two families of technique attack Equation 2.4 from opposite sides, summarised in Figure 6. **Compression** keeps one packet per reading and shrinks PL : general lossless coders such as Huffman and Lempel-Ziv-Welch adapted to sensor streams [12, 82–85] report 20 to 40% payload reduction but require dictionary or frequency-table state that a microcontroller with a few kilobytes of free RAM can only marginally afford [13, 86], and adaptive sampling reduces transmitted volume by skipping readings, which risks missing the threshold event the network exists to detect [87]. **Decomposition** takes the opposite path: it accepts more packets in exchange for each one being far shorter, splitting a reading across parallel channels rather than compressing it on one. Residue-based decomposition for wireless sensor data has been explored for storage reduction and for wireless sensor network lifetime [88–91], but, as Section 2.2(d) details, not for exploiting a multi-channel concentrator’s ability to demodulate several sub-channels *simultaneously*, which is the specific mechanism we study. LoRa-specific network-capacity work independently confirms that channel occupancy, not raw byte count, is the binding constraint on gateway-served node population [92–97], which is the same quantity Equation 2.5 fixes.



Compression shrinks the payload on one channel; decomposition trades packet count for per-packet occupancy across several. The literature has developed the left branch thoroughly and the right branch for storage and lifetime, but not yet for concurrent multi-channel reception.

Figure 6. Transmission-efficiency strategies for constrained IoT radios; we occupy the open leaf on the right.

Table 2 summarises representative results across both branches on the metric that matters for gateway capacity, per-packet airtime or its energy equivalent, rather than the compression ratio each source reports on its own terms.

Table 2. Representative low-power wide-area transmission-efficiency results

Study	Technique	Reported limitation
Li and Simonot-Lion [82]; Marcelloni and Vecchio [12]	Huffman/LZW sensor compression	Dictionary or frequency-table state exceeds the free RAM of the smallest microcontroller class; ratio degrades on low-entropy sensor streams
Kjellsson et al. [87]	Adaptive, event-driven sampling	Reduces volume, not per-packet occupancy; risks missing threshold events between samples
Vidhya and Brindha [88]	CRT residue image compression	Targets storage footprint, not transmission concurrency; single-channel model
Buttigieg et al. [89]; Pourbigharaz and Yassine [90]	RNS energy reduction in WSNs	Energy accounting only; no gateway-capacity or multi-channel reception model
Ye et al. [91]	RNS-based reliable WSN transmission	Reliability under loss, not concurrent channel exploitation

d) Residue Number Systems and the Chinese Remainder Theorem in

Communications

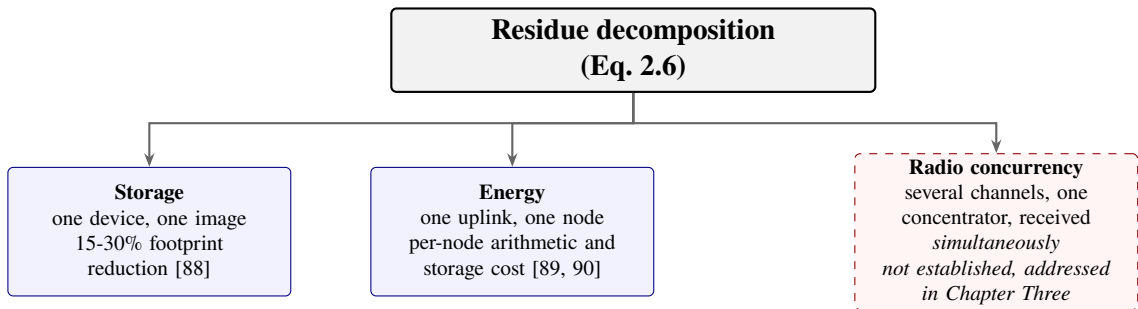
The theorem itself is unconditional. For pairwise coprime moduli $m_1, \dots, m_k$ and $M = \prod_{i=1}^k m_i$, every integer $x \in [0, M)$ is uniquely determined by its residue vector $(r_1, \dots, r_k)$, $r_i = x \bmod m_i$ [14, 37]. Garner's algorithm reconstructs x from the residues in closed form,

$$x = \left(\sum_{i=1}^k r_i M_i N_i \right) \bmod M, \quad M_i = \frac{M}{m_i}, \quad N_i = M_i^{-1} \bmod m_i, \quad (2.6)$$

exact because $M_i N_i \equiv 1 \pmod{m_i}$ by construction [14, 36], and computable in $O(k)$ modular multiplications regardless of the magnitude of x , which is the property that later work in residue number systems generalised into a programme for fast, carry-free, parallel arithmetic [15, 35]. A second property of Equation 2.6 bears on reliability rather than speed. Goldreich, Ron and Sudan formalised the error-tolerant case, showing that x remains recoverable from a strict subset of residues under bounded corruption, "Chinese remaindering with errors" [98], a result later systematised alongside subset-product reconstruction in computational number theory [38]. That subset-recovery property is a statement about *reconstruction*, not about *disclosure*: it is tempting to read a residue set as a threshold secret-sharing scheme, exact recovery from a complete or near-complete subset, negligible information from any one share in isolation, but the two constructions are not equivalent. A Shamir-style share is deliberately randomised so that any share short of the threshold is uniformly distributed independent of the secret [38]; a plain CRT residue is not. $r_i = x \bmod m_i$ is a deterministic function of x , so intercepting it narrows the field of possible values by a factor of roughly m_i rather than leaking nothing, and if the plaintext domain is itself small relative to m_i , a single residue can narrow that field to very little uncertainty indeed. Whether residue disclosure is negligible in practice therefore

depends on the ratio between the plaintext's domain and the moduli chosen, not on the reconstruction identity alone, and no prior work has established that ratio for any deployed system; Section 2.3(c) returns to this distinction, and Chapter One's hypothesis H5 treats it as open rather than established.

Two application communities have used Equation 2.6 for two different resources, and both stop short of a shared wireless channel. The first treats it as a *storage* problem: residue-based lossless image compression reports 15 to 30% footprint reduction with exact reconstruction [88]. The second treats it as an *energy* problem inside wireless sensor networks, converting a single wide-range reading into several narrow-range residues to shrink the arithmetic and storage cost per node [89–91]. Neither treats the residues as packets to be received concurrently on separate radio channels; both keep the decomposition on one device (compressing what is stored) or across one uplink (reducing what one node computes and sends), rather than exploiting a multi-channel concentrator's ability to demodulate several sub-channels *simultaneously*. To our knowledge, no published system applies residue decomposition to concurrent multi-channel radio reception, which is the gap Chapter Three's design addresses. Figure 7 lays the three resources side by side: two are established in the literature just surveyed, the third is open.



Both established applications keep the decomposition on one device or one uplink; neither exploits concurrent reception across a multi-channel concentrator, which is the resource this thesis spends the decomposition on.

Figure 7. The three resources residue decomposition has been applied to; radio concurrency remains open in the surveyed literature.

Residue number systems have a third, older home outside networking entirely: fast computer arithmetic and hardware design, where the absence of carry propagation across residue channels makes addition and multiplication in each channel independent of every other, at the cost of comparison and division becoming hard [15, 35]. That same independence gives residue codes a natural fault-tolerance reading, since a corrupted or missing channel affects only the value it encodes rather than propagating a carry error into its neighbours, which is part of why redundant residue codes have long been used for error detection in hardware [36, 37]. The parallelism this independence buys is architecturally the same kind of parallelism that the blockchain-scalability literature pursues by partitioning transaction load across shards or channels rather than processing it as one serial stream [18, 99]: both decompose a workload into independently processable parts and pay a bounded recombination cost to reassemble the result. We are not aware of published work that states this parallel explicitly or that applies residue decomposition to a blockchain-connected radio link; Section 2.3(d) returns to the blockchain side of that comparison once the scalability literature itself has been surveyed.

2.3. Literature Specific to the Topic

Where the foundations are general, the work below sits directly on the problem this thesis addresses, and is grouped by the property it tries to secure.

a) Blockchain and Distributed Ledger Technology in Agriculture and Supply Chains

Section 2.2(a) sets out the hash-chaining identity (Equation 2.1) that gives a replicated ledger its tamper evidence. Applied to agriculture, this tamper-evidence property has been used for supply-chain transparency [42, 100–106], farm-data

management [25, 107–110], and smart-contract automation of certification and payment logic [105, 111–113]. Cross-organisational data sharing has more recently been framed through self-sovereign identity, letting a farm operator authorise fine-grained access without handing a consortium a standing credential [114], and water-quality and irrigation-specific ledgers report similar tamper-evidence gains [25, 115–117]. The generic IoT-blockchain integration literature that agricultural systems draw their architecture from independently confirms the same design pattern: a lightweight device signs and forwards, a gateway or edge node validates and submits, and the ledger anchors a hash rather than the raw payload [104, 118–121].

Crop- and commodity-specific traceability deployments give this pattern concrete form. Salah et al. [44] traces soybean shipments end to end on Ethereum, anchoring custody transfers as they cross the supply chain; a comparable case study for ginger in Nepal reports the same design applied to a smallholder export crop [122], and a pilot study of blockchain for agriculture and food more broadly finds the technology’s actual value concentrated in dispute resolution and recall speed rather than in the blockchain mechanism’s novelty as such [123]. Reviews that step back from individual case studies converge on transparency and traceability as the two benefits every deployment claims, whatever the crop [124], and on collaborative trust and cross-party transparency as the property that justifies choosing a shared ledger over a conventional shared database in the first place [52]. A feasibility analysis of blockchain and IoT for African agricultural food supply chains [125] is the closest match in this corpus to our own deployment context: it names intermittent connectivity, low digital literacy and capital cost, rather than the blockchain mechanism itself, as the binding constraints on adoption, which is consistent with the constrained-hardware and field-deployment focus of Chapter Three’s methodology and reinforces why we evaluate on the low-cost, battery and

solar-powered hardware class that a smallholder deployment can actually afford, rather than on server-class infrastructure.

What this body of literature does not establish is where the tamper evidence begins. Almost every system surveyed authenticates at the gateway or the enterprise boundary and treats the sensor as an untrusted data source whose values are accepted once they arrive, not a principal whose identity is bound to the record it produced [25, 100, 126, 127]. Table 3 makes this concrete across a representative sample: the column that is consistently weak or absent is node-level, rather than gateway-level, identity binding, which is precisely the gap Chapter One’s problem statement (Section 1.2) identifies.

Table 3. Representative blockchain-in-agriculture systems by authentication boundary and evaluation setting

Study	Authentication boundary	Evaluation setting	Reported gap
Tian [42]; Kamilaris et al. [100]	Enterprise consortium	/ Conceptual, no field data	No per-sensor identity; throughput not characterised on constrained hardware
Zhang et al. [110]	Gateway	Simulation	Consensus cost assumed server-class hardware
Amraouy et al. [114]	Farm operator (self-sovereign identity)	Prototype	Access-sharing framework; no edge-hardware performance data
Reyna et al. [104]; Dorri et al. [118]	Generic IoT gateway	Survey architecture	/ Identifies the sensor-to-gateway trust gap without closing it

b) Edge-Resident Consensus and Constrained-Hardware Ledgers

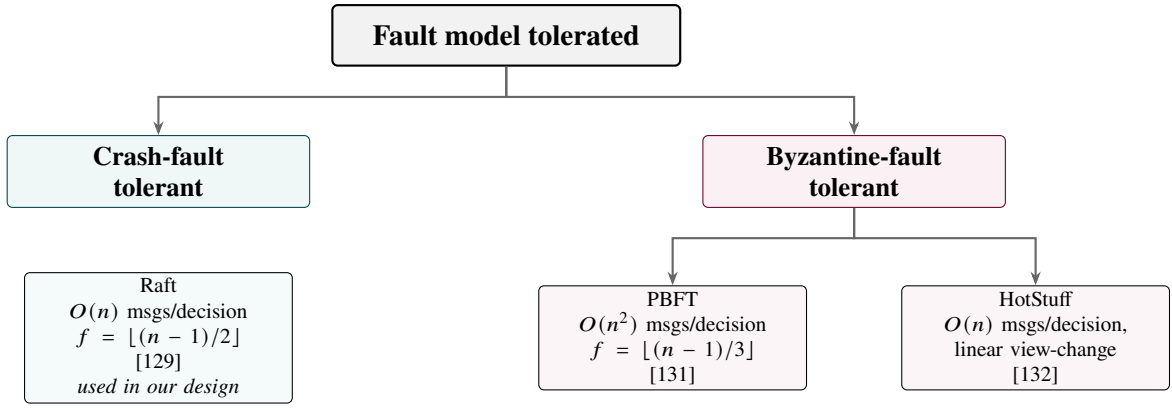
Running a permissioned ledger on gateway-class hardware raises a question the enterprise blockchain literature rarely poses: whether consensus can complete inside the timing budget of a real-time control loop, on a device with an order of magnitude less memory and Central Processing Unit (CPU) throughput than the server hardware that literature assumes [22, 23, 128]. Hyperledger Fabric’s execute-order-validate architecture separates transaction endorsement from ordering, letting an ordering service

reach agreement on transaction sequence rather than transaction content, and the Raft ordering service we use implements crash-fault-tolerant, not Byzantine-fault-tolerant, agreement [129, 130]. The distinction has a direct complexity consequence. Crash Fault Tolerance (CFT) protocols such as Raft reach agreement with $O(n)$ message exchanges per decision and tolerate $f = \lfloor (n - 1)/2 \rfloor$ stopped nodes, whereas classical Byzantine Fault Tolerance (BFT) protocols such as Practical Byzantine Fault Tolerance (PBFT) require $n \geq 3f + 1$ replicas, so they tolerate the smaller $f = \lfloor (n - 1)/3 \rfloor$, at $O(n^2)$ messages per decision, to withstand arbitrary, not merely crash, node behaviour [129, 131]. A five-node cluster therefore tolerates two stopped nodes under Raft but only one Byzantine node under PBFT, a smaller fault budget bought at a quadratically larger message cost. Linear-complexity BFT variants such as HotStuff close much of that gap for adversarial settings [132], and the state-machine-replication literature more broadly treats the CFT/BFT choice as a trade between message cost and the threat model tolerated [133]. Deployed on a consortium of trusted, physically secured gateways rather than an open or adversarial network, we adopt CFT for that reason, a choice the consensus literature supports for permissioned agricultural settings [24] but rarely evaluates on gateway-class ARM64 hardware in the field, as Table 4 shows.

Whichever protocol is chosen, its saturation behaviour under load is standard queueing theory. Treating packet or transaction arrivals as Poisson and service time as deterministic (an M/D/1 queue), mean waiting time before service is

$$W_q = \frac{\rho}{2(1 - \rho)} T_{\text{service}}, \quad \rho = \lambda T_{\text{service}}, \quad (2.7)$$

where λ is the arrival rate and T_{service} the deterministic service time, so that total latency $L = T_{\text{service}} + W_q$ is flat while $\rho \ll 1$ and diverges as $\rho \rightarrow 1$. The scalability literature for both LoRa gateways and blockchain orderers uses Equation 2.7, or its close relatives, to



Message complexity scales with the fault model tolerated, not with hardware class. CFT is adopted here because the deployment is a trusted, physically secured consortium; prior work rarely reports which choice was measured on ARM64 gateway hardware rather than assumed server-class hardware.

Figure 8. Consensus fault models and their message complexity, after [129, 131–133].

project a saturation point from a small deployment rather than requiring one at scale [23, 134–137], but, as with the consensus literature above, validation against real field load rather than synthetic benchmark load remains the exception rather than the rule. The Consistency, Availability, Partition tolerance (CAP) theorem underlies the same trade from a different angle: no distributed data store can simultaneously guarantee consistency and availability under network partition [138, 139], and permissioned ledgers on intermittently connected rural gateways sit squarely inside that trade-off, a point the practitioner literature on eventual, Atomicity, Consistency, Isolation, Durability (ACID)-relaxed consistency in distributed systems makes explicit [140].

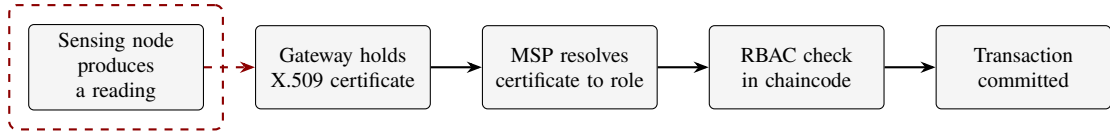
c) Identity, Access Control and Privacy-Preserving Data Processing

Two threads meet here: how a permissioned ledger decides who may act, and how it protects what a record discloses to parties who are entitled to see that it exists but not what it contains. On the first thread, RBAC enforced in chaincode is the standard mechanism, with permissions attached to roles rather than individual identities and a MSP resolving each transaction’s submitter to a role through its X.509 certificate [41, 141]. Flat RBAC

Table 4. Consensus and edge-ledger literature by hardware class evaluated

Study	Protocol	Hardware class	Reported gap
Ongaro and Ousterhout [129]	Raft (CFT)	Server-class (original)	Correctness and understandability proof; no constrained-hardware evaluation
Castro and Liskov [131]	PBFT (BFT)	Server-class	$O(n^2)$ messaging cost not evaluated against gateway bandwidth
Androulaki et al. [23]	Fabric pipeline	E-O-V Data-centre reference deployment	Throughput figures not reproduced on ARM64
Rahimi et al. [136]; Cui et al. [135]	Fabric + sensor / off-chain storage	Prototype, short-duration	No sustained field deployment; saturation point not field-validated

is the common case in this literature: a fixed set of roles, each with a fixed permission set, checked at the point a transaction is submitted. Administrative role hierarchies, in which a senior role inherits every permission of the roles beneath it in a partial order, are well established in access-control theory more broadly and are used to express exactly this kind of inheritance without duplicating permission assignments across roles [142]. What the Fabric access-control literature consistently shows, hierarchical or not, is that enforcement happens at the *gateway*, the device holding the X.509 certificate that signs the transaction, not at the individual sensing node whose reading the transaction carries [141, 143]. An improved blockchain-based authentication protocol for IoT network management pushes credential issuance closer to the device but still assumes a network manager mediates enrolment rather than the constrained node enrolling itself [143]. Certificate revocation latency under this model has been measured directly, with propagation delay across a small gossip-connected peer set on the order of minutes rather than seconds [142], a figure we reproduce independently in Chapter Four rather than merely assume. We develop a hierarchical role structure that closes the node-level identity gap this literature leaves open, formalised in Appendix B and implemented in Chapter Three, rather than presented here as part of the literature under review. Figure 9 places that gap on the enforcement pipeline this literature otherwise establishes.



identity gap

Enforcement in this literature begins at the gateway's certificate, hierarchical or not; the sensing node that produced the reading carries no credential of its own.

Figure 9. Where identity enforcement begins in the surveyed access-control literature, and the node-level gap it leaves.

On the second thread, privacy in a replicated ledger is structurally difficult: the same replication that gives every consortium member a tamper-evident copy of a record also gives every member a copy of whatever that record discloses. Off-chain storage with an on-chain hash anchor is the common mitigation, verifiable without being world-readable [144], and self-sovereign identity schemes extend the same principle to access itself, letting a data owner grant and revoke visibility without a standing credential [114]. Neither line of work, however, asks whether the transmission layer itself can contribute to confidentiality before a record ever reaches the ledger. Section 2.2(d) showed why that question is harder than it first looks: reconstruction tolerates a missing residue under Goldreich, Ron and Sudan's error-tolerant formulation [98], but a plain CRT residue is not a randomised secret share, so subset-recovery tolerance does not, by itself, establish subset-disclosure resistance [38]. What it establishes instead is a precise condition worth testing rather than assuming: disclosure from an intercepted residue is bounded by how far the plaintext's domain exceeds the missing moduli's product, so confidentiality is a property of the deployed moduli-to-domain ratio, not of the reconstruction identity in the abstract. Prior work has developed residue decomposition and threshold secret sharing in separate communities without stating that condition, which is exactly why Chapter One frames it as hypothesis H5 rather than a settled result: the rationale (Section 1.3) proposes residue decomposition as a candidate confidentiality

substrate, and H5 requires either a quantified leakage bound for the deployed moduli and sensor ranges, or an added randomisation step, before the substrate can be called established.

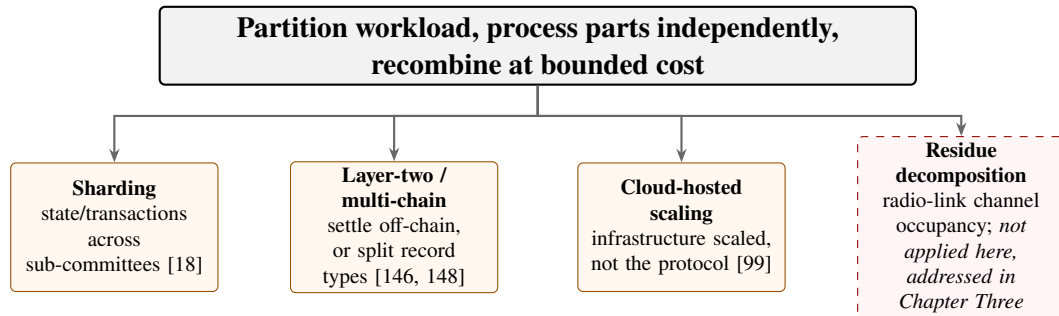
d) Blockchain Scalability and the Parallelism Argument

A substantial body of recent work attacks ledger throughput directly, and it matters here because the parallelisation argument we make for the radio link in Chapter Three has an established precedent in this literature: partition a workload, process the parts independently, and pay a bounded cost to recombine them.

Partitioning and sharding. Sharding partitions state and transaction load across sub-committees so that disjoint transactions commit concurrently. Recent treatments add adaptive assignment of accounts to shards [18] and survey the cost, chiefly cross-shard coordination, which grows with the fraction of transactions touching more than one partition [145]. Maiwada et al. [17] and Maiwada [16] analyse Bitcoin throughput under transaction-adjourning and related scheduling techniques, reporting the familiar result that raw block-level parallelism is bounded by the serialisation the consensus protocol imposes; an unrelated study of Bitcoin’s own Unspent Transaction Output (UTXO) transaction patterns reaches the same conclusion from the storage side, showing that throughput and storage growth trade against one another however the workload is scheduled [48]. Layer-two constructions move the work off chain entirely and settle in aggregate [146], hierarchical models have been used to characterise where the resulting systems saturate [147], and cloud-hosted blockchain frameworks pursue the same throughput goal by scaling the underlying infrastructure rather than the protocol, trading a capital cost for a scalability one [99]. Multi-chain architectures, keeping several independent ledgers for different record types and coordinating across them only

where required, are a further partitioning strategy outside the classic sharding literature, illustrated by a judicial case-management system that keeps case records and evidentiary records on separate chains [148].

The common structure of these results. Every one of these mechanisms decomposes a workload into parts that can be processed independently and recombines the parts afterwards, and in every case the recombination cost is what bounds the gain. That is structurally the same shape as a residue decomposition, where the recombination is Equation 2.6 and its cost is $O(k)$, known in advance and independent of the data, as Section 2.2(d) observes. None of the scalability studies surveyed here states that parallel explicitly, and none applies a residue decomposition to the workload it partitions; the common structure is, to our knowledge, an observation this review is making by reading the two literatures side by side, not a result either literature has published. Whether the same recombination-bounded-gain argument holds one layer down, applied to a radio link's channel occupancy rather than a ledger's transaction load, is a question Chapter Three's design takes up. Figure 10 sets the four instances against the one shared pattern.



Three instances of the pattern are established in the blockchain-scalability literature; the fourth, a residue decomposition recombined at the same $O(k)$ bounded cost, is the shape Chapter Three's design gives the radio link.

Figure 10. The partition-process-recombine pattern shared by blockchain scalability mechanisms, extended here to the radio link.

e) Residue Arithmetic Beyond Computer Arithmetic

The CRT literature outside computer arithmetic is small but shows the theorem migrating out of its traditional settings. Luo [19] surveys its modern algebraic forms, including statements over polynomial rings and group-theoretic settings, which establishes that the decomposition is not tied to integer congruences. Seethalakshmi and Spallone [149] develops an analogue for partitions. Of more direct relevance, Yavuz [20] uses CRT decomposition together with chaotic maps for reversible data hiding in encrypted images, which is the clearest indication among these sources that residues can carry a confidentiality property and not only an efficiency one. That result is suggestive for hypothesis H5 of Section 1.6, though it is established for a different medium and threat model and cannot be transferred without a fresh argument.

Oktaviansyah, Kurniadi, and Kusuma [150] combines the theorem with Hensel's lemma and the fundamental theorem of arithmetic for modular decomposition problems. None of this work applies the decomposition to wireless channel occupancy, which remains the gap identified in Section 2.2(d).

f) Permissioned Ledgers on Constrained Hardware

Evidence about Hyperledger Fabric outside the datacentre has begun to appear. Zulkarnain, Ramli, and Nordin [151] benchmarks Fabric across heterogeneous hardware for IoT workloads, and Pebriyanti and Suranegara [152] evaluates its scalability and resilience, both reporting that throughput falls well below datacentre figures while remaining usable for low-rate telemetry. Kwadzode [153] compares platforms under financial transaction models, which is a heavier workload than agricultural telemetry and therefore an upper bound on the cost.

Gong et al. [6] surveys edge-computing architectures for smart agriculture and

names communication reliability and multi-dimensional energy constraints as the two bottlenecks that most often defeat field deployments, which matches the framing of Section 2.2(c). Figure 11 reproduces that survey’s own framework, sensing, lightweight edge intelligence and edge-cloud collaboration feeding three agricultural scenarios; a permissioned ledger does not appear as a component at all. What these papers share with the earlier edge literature is that none couples the ledger to the transmission scheme: the radio and the ledger are treated as separate layers with separate budgets.

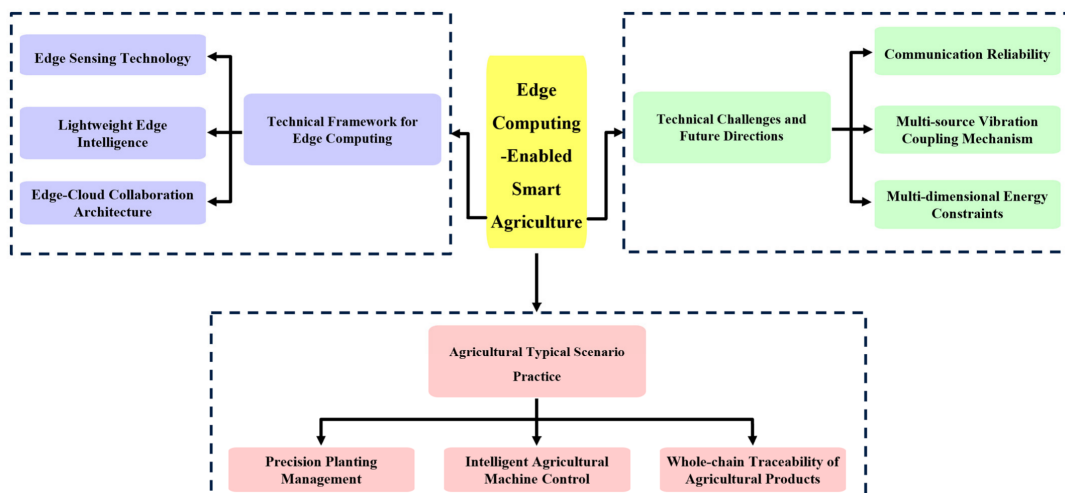


Figure 11. Edge computing framework for smart agriculture, reproduced under CC BY 4.0 from [6].

g) Identity, Privacy and Access Control in Adjacent Domains

The most developed treatment of identity and privacy over a permissioned ledger comes from electronic health records, where the regulatory pressure is greatest. Tang et al. [154] proposes an authentication scheme binding records to enrolled principals; Rajput et al. [155] adds emergency access control with after-the-fact auditability; Yousif, Alsaffar, and Ameen [156] and Hurdson and Kamaghe [157] address external sharing without disclosing the underlying record. Kormiltsyn et al. [158] treats the conflict that arises when personal and institutional records are integrated, which is structurally the

same problem as a farmer’s data entering a consortium ledger.

The agricultural instances are thinner but converge on the same requirement. Aliyu and Liu [25] addresses tampering in farm IoT data, and Safeer et al. [26] reports a production deployment in the Italian tomato chain. Aouedi et al. [7] and Sahu and Mazumdar [8] survey the IoT threat surface and agree that constrained endpoints remain the weakest link, which is the observation Section 1.2 builds on.

Two limits recur across this literature. Authentication is enforced where compute is available, that is at the gateway or the server, not at the sensing node; and confidentiality is obtained by withholding data from the ledger, which reintroduces a trusted party. Neither limit is addressed by a mechanism that operates on the reading in flight.

2.4. Research in Cognate Areas Relevant to the Topic

Three adjacent bodies of work inform our work without belonging to any of the themes surveyed above.

General edge computing and IoT security, beyond agriculture, establishes both the promise and the residual risk of pushing computation to the network edge. Edge processing improves response latency and keeps sensitive data local [159–161], and this holds across smart-city and vehicular deployments as much as agricultural ones, a distributed vehicular network architecture built on Merkle-tree-verified blocks being one such example [47]. The security side of the same literature catalogues IoT-specific attack surfaces, firmware injection, authentication bypass and physical tampering among them [118, 162–166], and a smaller literature addresses the specific hardware we deploy, cache and memory-region behaviour on the ESP32 family, relevant to firmware-level integrity rather than network protocol design [167–169].

Distributed and parallel computing beyond blockchain supplies the scaling vocabulary used in Section 2.3(b) and in Chapter Four: Amdahl’s law bounds the

speedup obtainable from parallelising a fixed-size workload by the fraction that remains sequential, a ceiling our strong-scaling analysis of gateway count against transaction throughput invokes directly [170]. Ongoing revocation-latency work in vehicular and other resource-constrained permissioned-blockchain settings similarly treats edge computing as the deployment target rather than the enterprise data centre [142], reinforcing the pattern Section 2.3(b) identifies: constrained-hardware evaluation is the literature’s exception rather than its default.

Applied hydrology and arid-region agronomy ground our water-efficiency claims outside the networking and ledger literature proper. Global freshwater withdrawal projections identify irrigation as the dominant consumptive use and the primary lever for closing the gap between supply and demand under population growth [171, 172], and Mediterranean and North African basin studies, including region-specific work on Tunisia [173–175], quantify the scale of the shortfall our field site sits inside. Crop-specific water-stress and quality studies for tomato and pepper cultivation, the two crops monitored in our deployment, supply the agronomic baselines against which Chapter Four’s yield and water-use results are compared [26, 176–184].

2.5. Critique of Current Literature and Summary of Known Results

Read across the threads surveyed above, the literature is strong on isolated mechanisms and weak on their combination and their field validation. Table 5 states this critique per thread in the same form our own contribution (Section 2.7) will answer it.

Three cross-cutting patterns follow from the table. First, every thread that reports a resource saving, transmission, storage, or processor cycles, reports it in isolation, with no accounting of what that saving could purchase elsewhere in the same pipeline; Section 1.3 of Chapter One turns this observation into our central rationale. Second, authentication and consensus evaluation both stop at the gateway, leaving the sensing node, the point at which

Table 5. Critique of current literature by thread, and what remains unestablished

Thread	What the literature establishes	What remains unestablished
Precision agriculture & IoT sensing (§2.2(b))	Sensor-driven scheduling reduces water use; decision-support benefit is replicated across regions	Data-integrity assumptions behind those results are rarely stated or measured; field sensing error budgets are under-reported
LPWAN transmission & energy (§2.2(c))	Compression reduces payload; residue decomposition reduces storage and per-node energy in isolation	No prior system uses residue decomposition for concurrent multi-channel <i>reception</i> , the mechanism that governs gateway capacity
RNS / CRT in communications (§2.2(d))	Reconstruction is exact and $O(k)$; subset recovery tolerates missing residues; parallel decomposition accelerates cryptographic verification	The processor-cycle application and the radio-channel application have not been connected; confidentiality implications of residue disclosure are unexploited
Blockchain / DLT in agriculture (§2.3(a))	Hash-chained, replicated ledgers give tamper evidence for agricultural and supply-chain records	Authentication almost universally terminates at the gateway or enterprise boundary, not the sensing node
Edge-resident consensus (§2.3(b))	CFT/BFT trade-off is well characterised in message complexity; CAP constrains any partitioned deployment	Constrained ARM64 gateway-class evaluation, especially under sustained field load rather than benchmark load, is the exception
Identity, access control & privacy (§2.3(c))	RBAC/MSP identity binding and role hierarchies are formalised and enforceable in chaincode	Node-level (not gateway-level) identity binding is largely absent; transmission-layer confidentiality is not treated as a design resource

falsification is cheapest, outside the trust boundary the literature actually measures. Third, laboratory and simulated evaluation dominates every thread; sustained field deployment, of the kind Chapter Three’s methodology and Chapter Four’s results describe, is reported by a minority of the studies reviewed here, and by none that combine transmission efficiency, edge consensus and access control in a single system.

2.6. Research Frontiers

Four questions are live in the literature discussed above and unresolved in it.

Whether decomposition can serve efficiency and confidentiality at once.

Residue representations are used for throughput in the ledger literature [18, 146] and, separately, for concealment in the image-security literature [20]. No work discussed here establishes the two properties for a single decomposition under one threat model.

How far identity can be pushed toward the sensor.

Every scheme surveyed in Section 2.3(g) terminates authentication at a device with meaningful compute. Whether a node with a few kilobytes of free memory can carry a credential strong enough to bind an individual reading, and at what airtime cost, is open.

What permissioned consensus actually costs in the field.

Benchmarks now exist for constrained hardware [151, 152], but they are laboratory measurements. Behaviour under mesh partition, thermal derating and seasonal power variation is reported nowhere in this body of work.

Where the binding constraint sits as deployments grow.

Equations 2.4 and 2.7 are sensitive to different quantities, so node-count and packet-capacity limits bind under different conditions. No validated model in prior work distinguishes them for multi-channel residue traffic.

2.7. The Contribution This Study Will Make to the Literature

Against the gaps summarised in Table 5, we contribute the following.

1. We connect the processor-cycle application of CRT parallel decomposition that the residue-arithmetic literature establishes (Section 2.2(d)) to a radio-channel application, showing that Equation 2.6 pays for a different resource, airtime rather than CPU cycles, when its parallel targets are a concentrator’s demodulation paths rather than a processor’s cores.
2. We extend gateway-terminated identity binding to the sensing node, using the hierarchical RBAC formalisation of Equations B.1 and B.2 (Appendix B) together with per-node credentials, so that a ledger record attests to the node that produced a reading rather than only to the gateway that submitted it.
3. We report CFT consensus performance measured on ARM64 gateway hardware under sustained field load rather than benchmark or simulated load, addressing the field-validation gap Table 4 identifies across the consensus literature.
4. We state, and Chapter Four tests as hypothesis H5, the condition under which residue decomposition can serve as a confidentiality substrate rather than merely a transmission optimisation: that disclosure from an intercepted residue is bounded by the ratio between the plaintext domain and the missing moduli’s product, a condition the subset-recovery results of [38, 98] make precise but do not by themselves establish for any deployed system, connecting the transmission and privacy threads that Section 2.3(c) shows have developed separately.
5. We report the combined system, transmission, ledger and access control, characterised together on one deployment, rather than each mechanism validated in

isolation, which Section 2.5 identifies as the pattern prior work does not yet show.

2.8. Partial Conclusion

The sensing literature has solved measurement and left trust untouched: the reading is believed because it was reported, and no mechanism attests that the value stored is the value measured [5, 34]. The transmission literature has correctly identified airtime as the binding constraint and attacks it by shortening payloads, which reduces total volume but not the per-packet occupancy that the duty-cycle and airtime relations of Equations 2.5 and 2.4 are actually sensitive to [12, 13]. Residue arithmetic supplies a decomposition whose parallelism has been exploited in silicon and, recently, in ledger throughput, but never on the air [16, 19, 35]. The ledger literature supplies tamper evidence and an identity layer, characterised on hardware that agricultural deployments do not have [22, 151]. The privacy literature obtains confidentiality by withholding, which forfeits the verifiability that motivated the ledger in the first place [156, 158].

Taken together these five leave one composite question, which we address in the chapters that follow: whether a decomposition adopted to reduce per-packet occupancy can simultaneously carry node-level identity and bound disclosure, and whether the resulting system holds up on the hardware and under the conditions that field deployment actually imposes.

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APPENDIX A

WORKED RECONSTRUCTION EXAMPLE FOR THE CHINESE REMAINDER THEOREM

Section 2.2(d) states Garner's reconstruction identity (Equation 2.6) in general form. This appendix carries one complete numeric instance through every step, including the modular-inverse computations that Chapter Two states as results, so that the arithmetic underlying Chapter Three's implementation is checkable by hand.

Take the moduli set $\{97, 101, 103\}$, pairwise coprime, with product

$$M = 97 \times 101 \times 103 = 1,009,091.$$

A sensor reading of 25.30°C , scaled by 100 to keep the value integral, gives $x = 2530$.

Its three residues are

$$r_1 = 2530 \bmod 97 = 8,$$

$$r_2 = 2530 \bmod 101 = 5,$$

$$r_3 = 2530 \bmod 103 = 58.$$

Step 1: partial products. For each modulus m_i , $M_i = M/m_i$:

$$M_1 = 101 \times 103 = 10,403,$$

$$M_2 = 97 \times 103 = 9,991,$$

$$M_3 = 97 \times 101 = 9,797.$$

Step 2: modular inverses. Each N_i satisfies $M_i N_i \equiv 1 \pmod{m_i}$. Reducing M_i modulo m_i first keeps the inverse search small. For $m_1 = 97$:

$$M_1 \bmod 97 = 10,403 \bmod 97 = 24,$$

$$24 \times 93 = 2,232 = 23 \times 97 + 1 \Rightarrow N_1 = 93.$$

For $m_2 = 101$:

$$M_2 \bmod 101 = 9,991 \bmod 101 = 93,$$

$$93 \times 63 = 5,859 = 58 \times 101 + 1 \Rightarrow N_2 = 63.$$

For $m_3 = 103$:

$$M_3 \bmod 103 = 9,797 \bmod 103 = 12,$$

$$12 \times 43 = 516 = 5 \times 103 + 1 \Rightarrow N_3 = 43.$$

Each inverse is found by inspection here because the moduli are small; the gateway firmware computes the same quantity with the extended Euclidean algorithm, $O(\log m_i)$ per modulus, a sub-millisecond cost at this moduli size (Chapter Three reports measured decomposition and reconstruction timings).

Step 3: reconstruction. Substituting into Equation 2.6:

$$\begin{aligned} x &= (r_1 M_1 N_1 + r_2 M_2 N_2 + r_3 M_3 N_3) \bmod M \\ &= (8 \times 10,403 \times 93 + 5 \times 9,991 \times 63 + 58 \times 9,797 \times 43) \bmod 1,009,091 \\ &= 2,530, \end{aligned}$$

exact, with zero reconstruction error, recovering 25.30°C after descaling. Losing any one residue, say r_3 , leaves a unique solution modulo $m_1m_2 = 9,797$, which happens to be exact for *this particular* value because $2,530 < 9,797$.

Why that qualifier matters. It does not generalise to every encoded value the firmware produces. The single-sensor value worked above fits comfortably under any two-modulus product, but the deployed encoding in `esp32/main/main.c` packs a soil-moisture reading and a temperature bucket into one integer, $x = \text{soil} \times 201 + \text{temp_bucket}$, precisely so that the full three-modulus budget of Section 2.2(d) is used; with soil scaled to $[0, 4095]$, x reaches up to 823,295, which requires all three moduli, since no pair's product exceeds 10,403. The gateway's decode routine (`gateway/crt_decode.py`) accepts any two residues and returns the smallest non-negative solution modulo their product, documenting that this "matches gateway reassembly for sensor values constrained below the product of the received moduli" rather than guaranteeing correctness in general; for a combined value at or above that product, two-residue decoding returns a different, aliased value rather than signalling failure. This is why the subset-recovery figure the source article reports is a measured 64.1% against a 66.7% theoretical figure for two-of-three recovery, not the 100% that would hold if every encoded value stayed under every two-modulus product; when Chapter Three's own data-recovery table is written, it should carry the same measured figure rather than a general bound. The subset-recovery property of Section 2.2(d) is real and exact for values small enough relative to the moduli in play; it is not, as deployed here, a blanket guarantee, and Chapter Three's treatment of loss resilience should state the recovery rate as measured rather than imply the general bound applies unconditionally.

APPENDIX B

HIERARCHICAL ROLE-BASED ACCESS CONTROL FORMALISATION AND PERMISSION MATRIX

Chapter Two’s review of the access-control literature (Section 2.3(c)) finds RBAC enforced in chaincode a well-established mechanism, generally terminating at the gateway, but no published, deployed formalisation of a *hierarchical* role structure with zone and temporal constraints folded into the grant decision itself. This appendix states the formalisation we develop to close that gap and that Chapter Three’s chaincode implements, so that it is available for reference wherever the main chapters invoke it, without presenting our own model as part of the literature under review.

We extend flat RBAC with a role hierarchy expressed as a partial order $\preceq$ over the role set R , where $r_i \preceq r_j$ iff $\text{permissions}(r_i) \subseteq \text{permissions}(r_j)$, so that a user u ’s effective permission set in a session is the union across every role dominated by an active role,

$$\text{eff}(u) = \bigcup_{r \in \text{active}(u)} \bigcup_{r' \preceq r} pa(r'), \quad (\text{B.1})$$

with $pa(r)$ the permissions directly assigned to role r . An access request (u, op, res, t) is then granted only if an active, non-expired role of u carries the requested permission *and* the requester’s zone matches the resource’s zone or the requester holds an administrator-issued cross-zone token,

$$\exists r \in \text{active}(u, t) : (op, res) \in \text{eff}(r) \wedge (\text{zone}(u) = \text{zone}(res) \vee \text{czt}(u, \text{zone}(res), t)), \quad (\text{B.2})$$

which folds zone and temporal constraints directly into the access decision rather than leaving them to an external policy layer. This follows the general tradition of administrative role hierarchies in access-control theory [142] and shares its enforcement point, the

chaincode’s client-identity check, with the wider Fabric access-control literature [141], but authenticates at the individual sensing node rather than only at the gateway holding the X.509 certificate that signs the transaction. Figure 12 shows the resulting enrolment chain, root certificate authority down through gateways to individual sensing nodes.

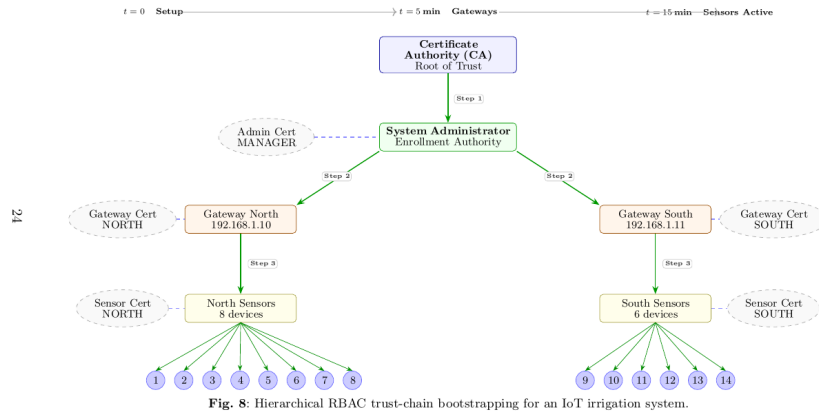


Figure 12. Hierarchical enrolment chain for the seven-role HRBAC model of Equations B.1 and B.2.

Table 6 reproduces the permission assignment $pa(r)$ for every role and resource Chapter Three’s chaincode governs, so that the effective-permission union in Equation B.1 can be checked against a concrete instance.

Table 6. Permission matrix for agricultural resources across the seven HRBAC roles.
own = own zone only, *zone* = assigned zone, *all* = all zones, N/A = no access.

Resource	Sensor	Gateway	Farmer	Agron.	Cert.	Supply	Admin
Sensor data	own	zone	all	all	all	N/A	all
Irrigation control	N/A	zone	full	N/A	N/A	N/A	full
Audit logs	N/A	N/A	N/A	N/A	read	N/A	full
Role assignments	N/A	N/A	N/A	N/A	N/A	N/A	full
Cross-zone tokens	N/A	N/A	N/A	N/A	N/A	N/A	issue

Note: Agron. = Agronomist, Cert. = Certification Body, Supply = Supply Chain Partner.

Two rows are worth reading against Equation B.2 directly. **Sensor data** is the

only resource every role can reach in some capacity, own-zone write for the sensor itself, zone-scoped read for the gateway, and unrestricted read for the roles that sit above FARMER in the partial order above; this is what makes the zone predicate $\text{zone}(u) = \text{zone}(res)$ load-bearing rather than redundant; without it, every role above Gateway would read every zone's sensor data by default. **Role assignments** and **cross-zone tokens** are reachable only by ADMIN, which is what closes Equation B.2's $\text{czt}(u, \text{zone}(res), t)$ predicate: a cross-zone token can only be minted by the one role whose own access is already unrestricted, so the token mechanism widens a requester's reach without ever widening the set of principals who can create that widening.